Stoichiometry

Mole ratios and mass calculations from balanced equations

Name _____ Class _____ Date _____

DATA

H = 1.008 g/mol C = 12.01 g/mol O = 16.00 g/mol Mg = 24.31 g/mol Cl = 35.45 g/mol

K = 39.10 g/mol Fe = 55.85 g/mol

Road map: grams → moles → (mole ratio) → moles → grams. Check that each equation is balanced.

A. Mole ratios

1. For $4\,Al + 3\,O_2 \longrightarrow 2\,Al_2O_3$, write the mole ratio of (a) O_2 to Al and (b) Al_2O_3 to Al.

2. Using the same equation, how many moles of O_2 react with 6.0 mol of aluminium?

B. Mass calculations

3. Magnesium burns: $2\,Mg + O_2 \longrightarrow 2\,MgO$. What mass of magnesium oxide forms from 12.15 g of magnesium?

4. Potassium chlorate decomposes on heating: $2\,KClO_3 \longrightarrow 2\,KCl + 3\,O_2$. What mass of oxygen forms from 24.5 g of $KClO_3$?

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5. Methane burns: $\mathrm{CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O}$. What mass of carbon dioxide forms from 8.02 g of methane?

6. Propane burns: $\mathrm{C_3H_8 + 5\,O_2 \longrightarrow 3\,CO_2 + 4\,H_2O}$. What mass of water forms from 11.0 g of propane?

C. Challenge

7. Iron is made from iron(III) oxide and carbon monoxide: $\mathrm{Fe_2O_3 + CO \longrightarrow Fe + CO_2}$ (unbalanced).

(a) Balance the equation. (b) What mass of iron can be made from 100.0 g of $\mathrm{Fe_2O_3}$?

8. A student says: "In $\mathrm{2\,Mg + O_2 \longrightarrow 2\,MgO}$, 2 g of magnesium reacts with 1 g of oxygen." Explain the mistake.

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